

Field Equations for the Causal Diamond Framework: Einstein's Equations with an Entropy-Determined Source

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Abstract

The causal diamond framework derives the MOND acceleration constant and interpolation function from the entropy structure of the observer's causal diamond in Λ CDM, but has lacked field equations. Here we show that the framework admits a simple relativistic formulation: the standard Einstein field equations with the baryonic stress-energy tensor divided by the entropy-balance interpolation function, $G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu} / \mu(|\nabla\Phi|/a_0)$. The equation structure is a known relativistic extension of AQUAL (Bekenstein & Milgrom 1984); what is new is not the structure but its content: both μ and a_0 are derived from the causal diamond, not input as free functions and parameters. The function μ is not a new dynamical field; it is an algebraic function of the gravitational field with no propagating degrees of freedom, no additional initial conditions, and no stability issues. The formulation automatically satisfies five consistency requirements: (1) Newtonian limit, (2) MOND limit, (3) $c_{\text{gw}} = c$, (4) lensing equals dynamics, and (5) total energy-momentum conservation. Equivalently, the framework is standard GR with an effective source $T_{\mu\nu}^{\text{total}} = T_{\mu\nu} / \mu$, where the phantom component $T_{\mu\nu}(1/\mu - 1)$ is the entropy deficit's gravitational signature.

1. Motivation

In companion papers [1–7], we derived the MOND acceleration constant $a_0 = c^2/(2\pi r_w)$ and the interpolation function $\mu(x) = kx/(1+kx)$ from the entropy structure of the cosmological causal diamond, showed that the gravitational wave speed is $c_{\text{gw}} = c$ [5], that lensing tracks dynamics [7], and that MOND effects vanish in the early universe [7]. These results were obtained without field equations. The framework operated at the level of an entropy balance that produces $\mu(x)$, without specifying how μ enters the relativistic gravitational equations.

This paper provides the missing field equations. The result is remarkably simple: the standard Einstein equations with a modified source.

2. The Field Equations

2.1 Standard Einstein equations

In general relativity, the Einstein field equations relate spacetime geometry to matter content:

$$G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu} \quad (1)$$

2.2 The entropy-modified field equations

In the causal diamond framework, the entropy balance reduces the fraction of degrees of freedom available for Newtonian gravity to $\mu(x)$, where $x = |\nabla\Phi|/a_0$. The effective gravitational coupling is $G_{\text{eff}} =$

G/μ . The field equations become:

$$G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu} / \mu (|\nabla\Phi|/a_0) \quad (2)$$

with the constraint equations:

$$\mu(x) = kx/(1+kx), \quad k \approx 1, \quad a_0 = 2c^2/(3\pi d_{\text{eh}}) \quad (3)$$

Equations (2) and (3) form a closed system. The function μ is determined algebraically by the gravitational field; it is not an independent dynamical variable and has no equation of motion. No new fields are introduced. The gravitational action remains the standard Einstein-Hilbert action. Only the source side of the equation is modified.

2.3 Relation to AQUAL and prior formulations

The structure of Eq. (2)—Einstein's equations with an effective coupling G/μ —is not new. In the Newtonian limit, it reduces to the modified Poisson equation $\nabla \cdot [\mu(|\nabla\Phi|/a_0) \nabla\Phi] = 4\pi G\rho$, which is the AQUAL formulation of MOND [8]. Milgrom [9] considered the possibility of a position-dependent effective coupling $G_{\text{eff}} = G/\mu$. Hossenfelder [10] explored a covariant version of Verlinde's emergent gravity with similar goals.

What is new in the present work is not the equation structure but its content. In AQUAL, $\mu(x)$ is a free function chosen to fit rotation curve data, and a_0 is a free parameter measured from observations. In the causal diamond framework, $\mu = kx/(1+kx)$ is derived from the entropy balance [2] and $a_0 = c^2/(2\pi r_w)$ is derived from the causal diamond waist [1]. The equation structure is known; the content is derived. This is the distinction between a phenomenological framework (AQUAL) and a physical one (the causal diamond).

2.4 Equivalent formulation: effective total source

Equation (2) can be rewritten as standard Einstein equations with an effective source:

$$G_{\mu\nu} = (8\pi G/c^4) T_{\mu\nu}^{\text{total}} \quad (4)$$

where

$$T_{\mu\nu}^{\text{total}} = T_{\mu\nu} / \mu = T_{\mu\nu} + T_{\mu\nu} (1/\mu - 1) \quad (5)$$

The second term, $T_{\mu\nu}^{\text{phantom}} = T_{\mu\nu} (1/\mu - 1)$, is the gravitational signature of the entropy deficit. It is what observers interpret as “dark matter.” In this formulation, the framework is literally standard GR with a specific prescription for computing the effective source from the baryonic matter distribution.

3. The Phantom Component in the Weak-Field Limit

In the weak-field, spherically symmetric limit (a galaxy with baryonic mass M), the Newtonian acceleration is $g_N = GM/r^2$. The entropy-deficit phantom mass enclosed within radius r is:

$$M_{\text{phantom}}(r) = M \times (1/\mu - 1) \quad (6)$$

For $\mu = kx/(1+kx)$ with $k = 1$ and $x = g_N/a_0$:

$$1/\mu - 1 = a_0/g_N = a_0 r^2/(GM) \quad (7)$$

The phantom mass is $M_{\text{phantom}}(r) = a_0 r^2/G$, and the phantom density is:

$$\rho_{\text{phantom}}(r) = a_0/(2\pi Gr) \quad (8)$$

This is the isothermal sphere profile $\rho \propto 1/r$, producing flat rotation curves with asymptotic velocity:

$$v^4 = GMa_0 \quad (9)$$

which is the baryonic Tully-Fisher relation [11], recovered automatically.

4. Consistency Checks

Test	Requirement	Result
Newtonian limit	$\mu \rightarrow 1$ at high acceleration	$G_{\mu\nu} = (8\pi G/c^2)T$ ✓
MOND limit	$\mu \rightarrow kx$ at low acceleration	Tully-Fisher recovered ✓
GW speed	$c_{\text{gw}} = c$	Vacuum: $T=0$, $1/\mu \times 0=0$ ✓
Lensing	Lensing = dynamics	Effective source curves spacetime ✓
Conservation	$\nabla_\mu T^{\text{total}} = 0$	Bianchi identity satisfied ✓

Table 1. Five consistency checks for the field equations. All are satisfied.

4.1 Gravitational wave speed

For gravitational wave propagation, we linearize around a vacuum background ($T_{\mu\nu} = 0$). The right-hand side of Eq. (2) vanishes identically, regardless of μ . The linearized vacuum equations are standard, and gravitational waves propagate at c . The $1/\mu$ factor multiplies the source, not the geometric side of the equation.

4.2 Conservation law

The Bianchi identity requires $\nabla_\mu G^{\mu\nu} = 0$, which implies $\nabla_\mu (T^{\mu\nu}/\mu) = 0$ —conservation of the total stress-energy. The baryonic stress-energy alone is not independently conserved:

$$\nabla_\mu T^{\mu\nu} = (T^{\mu\nu}/\mu) \nabla_\mu \mu \quad (10)$$

The right-hand side represents energy-momentum exchange between the baryonic and phantom sectors as matter moves through regions of varying μ . This feature is not unique to the causal diamond formulation; it is generic to all MOND theories with position-dependent effective gravitational coupling, including AQUAL [8]. In the high-acceleration regime ($\mu \approx 1$), the exchange vanishes and baryonic conservation is recovered.

5. What the Framework Is and Is Not

5.1 What it is

The causal diamond field equations (2)–(3) are standard Einstein equations with an algebraic prescription for computing the effective source from baryonic matter. The result is a theory with:

- No new dynamical fields—unlike TeVeS [12], which adds scalar and vector fields, and AeST [13], which adds a dynamical scalar. The algebraic nature of μ means no new propagating degrees of freedom, no additional initial conditions, and no stability issues from scalar field dynamics.
- No modification of the gravitational action (Einstein-Hilbert)
- Zero free parameters (a_0 and μ both derived)

5.2 What it is not

The formulation is not a complete fundamental theory. Several limitations apply.

First, Eq. (2) uses $|\nabla\Phi|$, which is a Newtonian concept. A fully covariant replacement requires specifying how to extract the “acceleration” from the spacetime metric in general. In static, spherically symmetric spacetimes, $|\nabla\Phi|$ maps to the norm of the four-acceleration of static observers. For general spacetimes (e.g., cosmological perturbations), a covariant definition of x is needed. This is the principal obstacle to extending the framework to CMB and structure formation calculations, and we leave it to future work.

Second, the field equations inherit the entropy normalization ambiguity from Ref. [4]. The value of k depends on the volume-law entropy density convention. The structure of Eq. (2) and the functional form of μ are independent of this choice.

Third, the conservation equation (10) implies energy exchange between the baryonic and phantom sectors. Whether this exchange has observable consequences beyond the MOND-modified dynamics (e.g., anomalous heating in low-acceleration environments) has not been computed.

6. Comparison with Other Relativistic MOND Theories

Property	AQUAL	TeVes	AeST	Causal Diamond
New fields	None (NR)	Scalar+vector	Scalar+vector	None
μ dynamics	Algebraic	Field eq.	Field eq.	Algebraic
a_{eff} origin	Input	Input	Input	Derived
$\mu(x)$ origin	Input	Input	Input	Derived
Relativistic	No	Yes	Yes	Yes (weak field)
$c_{\text{gw}} = c$	N/A	No (killed)	Yes (designed)	Yes (automatic)
Free parameters	2+	3+	3+	0

Table 2. Comparison of MOND frameworks. AQUAL is non-relativistic (NR). The causal diamond formulation shares AQUAL’s algebraic μ structure but provides derived content and a relativistic (weak-field) extension.

7. Outlook

The field equations open the door to three calculations that the framework could not previously attempt:

Galaxy clusters. Standard MOND underpredicts cluster masses by a factor of ~ 2 [14]. The field equations allow the phantom mass to be computed for extended, non-spherical mass distributions, potentially closing or reducing this gap.

Cosmological perturbations. With a covariant generalization of x , the field equations can be linearized around an FLRW background to compute the matter power spectrum and CMB anisotropies. This is the most important open calculation.

Structure formation. The phantom density $\rho_{\text{phantom}} \propto 1/r$ provides a gravitational potential well that could seed structure formation. Whether the growth rate matches observations is computable from the field equations.

8. Conclusion

The causal diamond framework admits a simple relativistic formulation: Einstein’s equations with the baryonic stress-energy divided by the entropy-balance interpolation function μ . The equation structure is a relativistic extension of AQUAL [8]; what is new is the derived content of μ and a_0 from the causal diamond. The function μ is algebraically determined by the gravitational field—it introduces no new propagating degrees of freedom.

Equivalently, the framework is standard general relativity with an effective source $T^{\text{total}} = T_{\text{baryonic}}/\mu$. The phantom component $T(1/\mu - 1)$ is the entropy deficit—the gravitational signature of dark energy’s consumption of degrees of freedom in the observer’s causal diamond. This is what observers have been calling dark matter.

The field equations pass five consistency checks and open the door to cluster-scale calculations, cosmological perturbation theory, and structure formation—the three tests that will determine whether the causal diamond framework is a correct description of gravity or an elegant approximation that breaks at larger scales.

References

- [1] D. Nguyen, “The Geometric Origin of the MOND Acceleration Constant: The Causal Diamond Waist in Λ CDM,” Zenodo (2026).
- [2] D. Nguyen, “Deriving the MOND Interpolation Function: Area-Law vs Volume-Law Entropy in the Causal Diamond,” Zenodo (2026).
- [3] D. Nguyen, “Erratum: The Physical Event Horizon as the Correct Entropy Normalization,” Zenodo (2026).
- [4] D. Nguyen, “The Volume-Law Entropy Normalization: What Can and Cannot Be Derived Without Quantum Gravity,” Zenodo (2026).
- [5] D. Nguyen, “Gravitational Wave Speed in the Causal Diamond Framework,” Zenodo (2026).
- [6] D. Nguyen, “The Cosmological Constant as a Sixth Power of the MOND Scale,” Zenodo (2026).
- [7] D. Nguyen, “Gravitational Lensing and Early-Universe Behavior in the Causal Diamond Framework,” Zenodo (2026).
- [8] J. D. Bekenstein and M. Milgrom, “Does the missing mass problem signal the breakdown of Newtonian gravity?,” *Astrophys. J.* 286, 7 (1984).
- [9] M. Milgrom, “Modified dynamics as an alternative to dark matter,” *Ann. Phys.* 229, 384 (1994).
- [10] S. Hossenfelder, “A Covariant Version of Verlinde’s Emergent Gravity,” *Phys. Rev. D* 95, 124018 (2017). arXiv:1703.01415.
- [11] S. S. McGaugh, “The Baryonic Tully-Fisher Relation,” *Astrophys. J.* 632, 859 (2005).
- [12] J. D. Bekenstein, *Phys. Rev. D* 70, 083509 (2004).
- [13] C. Skordis and T. Zlotnik, *Phys. Rev. Lett.* 127, 161302 (2021).
- [14] R. H. Sanders, *Mon. Not. R. Astron. Soc.* 342, 901 (2003).